

Electrochemistry: Formula Sheet

Class 12 Chemistry – Unit 2 Complete Revision Formulas

1 Electrochemical Cells & Cell Potential

- Standard Cell Electromotive Force (E_{cell}°):

$$E_{\text{cell}}^{\circ} = E_{\text{right}}^{\circ} - E_{\text{left}}^{\circ} = E_{\text{cathode}}^{\circ} - E_{\text{anode}}^{\circ}$$

Where $E_{\text{cathode}}^{\circ}$ and E_{anode}° are standard reduction potentials ($E_{\text{SHE}}^{\circ} = 0.00 \text{ V}$).

- Standard Gibbs Free Energy ($\Delta_r G^{\circ}$) & Equilibrium Constant (K_c):

$$\Delta_r G^{\circ} = -n F E_{\text{cell}}^{\circ}$$

$$\Delta_r G^{\circ} = -R T \ln K_c = -2.303 R T \log K_c$$

$$E_{\text{cell}}^{\circ} = \frac{2.303 R T}{n F} \log K_c = \frac{0.059}{n} \log K_c \quad (\text{at } T = 298 \text{ K})$$

Where n is number of moles of electrons transferred, and $F = 96487 \text{ C} \cdot \text{mol}^{-1} \approx 96500 \text{ C} \cdot \text{mol}^{-1}$.

2 Nernst Equation

- Single Electrode Reduction Potential ($M^{n+} + ne^{-} \rightarrow M$):

$$E_{(M^{n+}/M)} = E_{(M^{n+}/M)}^{\circ} - \frac{R T}{n F} \ln \left(\frac{1}{[M^{n+}]} \right) = E_{(M^{n+}/M)}^{\circ} - \frac{0.059}{n} \log \left(\frac{1}{[M^{n+}]} \right)$$

- General Cell Reaction ($aA + bB \xrightarrow{ne^{-}} cC + dD$):

$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{R T}{n F} \ln Q_c = E_{\text{cell}}^{\circ} - \frac{0.059}{n} \log \left(\frac{[C]^c [D]^d}{[A]^a [B]^b} \right) \quad (\text{at } 298 \text{ K})$$

For Daniell Cell: $E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.059}{2} \log \left(\frac{[\text{Zn}^{2+}]}{[\text{Cu}^{2+}]} \right)$.

3 Conductance, Conductivity & Molar Conductivity

- Resistance (R) and Conductance (G):

$$R = \rho \left(\frac{l}{A} \right), \quad G = \frac{1}{R} = \kappa \left(\frac{A}{l} \right) \quad [\text{Units: } R \text{ in } \Omega, G \text{ in Siemens (S) or } \Omega^{-1}]$$

- Cell Constant (G^*) and Conductivity (κ):

$$G^* = \frac{l}{A} = R \cdot \kappa \implies \kappa = \frac{G^*}{R} = G \cdot G^* \quad [\text{Unit: } S \cdot \text{cm}^{-1} \text{ or } S \cdot \text{m}^{-1}]$$

- Molar Conductivity (Λ_m):

$$\Lambda_m = \frac{\kappa}{c} = \frac{\kappa (\text{in } S \cdot \text{m}^{-1})}{1000 \times M} \quad [\text{Unit: } S \cdot \text{m}^2 \text{mol}^{-1}]$$
$$\Lambda_m = \frac{\kappa (\text{in } S \cdot \text{cm}^{-1}) \times 1000}{M (\text{in mol} \cdot \text{L}^{-1})} \quad [\text{Unit: } S \cdot \text{cm}^2 \text{mol}^{-1}]$$

- Debye-Hückel-Onsager Equation (Strong Electrolytes):

$$\Lambda_m = \Lambda_m^\circ - A c^{1/2}$$

Where Λ_m° is limiting molar conductivity at infinite dilution and A is a constant.

4 Kohlrausch's Law & Weak Electrolyte Dissociation

- Kohlrausch's Law of Independent Migration of Ions:

$$\Lambda_m^\circ = \nu_+ \lambda_+^\circ + \nu_- \lambda_-^\circ$$

Where ν_+ and ν_- are numbers of cations and anions produced per formula unit.

- Degree of Dissociation (α) and Dissociation Constant (K_a):

$$\alpha = \frac{\Lambda_m}{\Lambda_m^\circ} \implies K_a = \frac{c \alpha^2}{1 - \alpha} = \frac{c \Lambda_m^2}{\Lambda_m^\circ (\Lambda_m^\circ - \Lambda_m)}$$

5 Faraday's Laws of Electrolysis

- **First Law of Electrolysis:**

$$m = Z Q = Z I t = \left(\frac{M_{\text{atomic}}}{n F} \right) I t$$

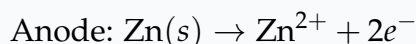
Where m is mass deposited [g], Z is electrochemical equivalent, I is current [A], and t is time [s].

- **Second Law of Electrolysis:**

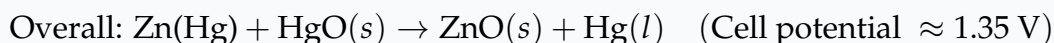
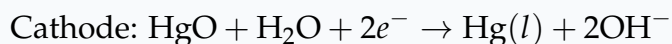
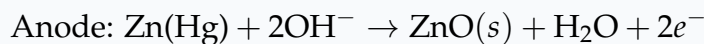
$$\frac{m_1}{m_2} = \frac{E_1}{E_2} \quad \left(E = \frac{\text{Atomic Mass}}{n} = \text{Equivalent Weight} \right)$$

6 Batteries, Fuel Cells & Corrosion Chemistry

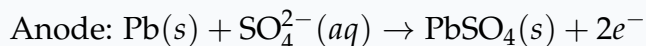
- **Dry Cell (Leclanché Cell):**



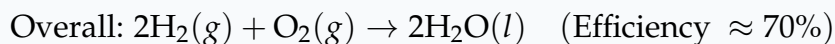
- **Mercury Button Cell:**



- **Lead-Acid Storage Battery (Discharge / Charge Cycle):**



- **Hydrogen-Oxygen (H₂-O₂) Fuel Cell:**



- **Corrosion / Rusting of Iron:**

Anodic site: $2\text{Fe}(s) \rightarrow 2\text{Fe}^{2+} + 4e^-$ ($E^\circ = -0.44 \text{ V}$)

Cathodic site: $\text{O}_2(g) + 4\text{H}^+(aq) + 4e^- \rightarrow 2\text{H}_2\text{O}(l)$ ($E^\circ = 1.23 \text{ V}$)

Overall cell: $2\text{Fe}(s) + \text{O}_2(g) + 4\text{H}^+(aq) \rightarrow 2\text{Fe}^{2+}(aq) + 2\text{H}_2\text{O}(l)$ ($E^\circ_{\text{cell}} = 1.67 \text{ V}$)

Rust Formation: $2\text{Fe}^{2+} + 2\text{H}_2\text{O} + \frac{1}{2}\text{O}_2 \rightarrow \text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O} + 4\text{H}^+$